

The Growth of Executive Constraints

LLM Pipeline Codebook

June 26, 2026

This codebook documents every column that appears in the pipeline output files (*.csv / *.json).

Coding principle. All indicators code *de facto* (actual) power, not de jure (formal) arrangements.

Default run settings. The pipeline is run with `--verify self_consistency --n-samples 2`, which makes 3 total LLM calls per indicator (1 initial + 2 additional samples). This produces per-indicator columns `_SC1`, `_SC2`, `_SC3`, plus `_verified` (majority vote), `_agreement`, and `_uncertainty`. If `--n-samples 1` is used, only `_SC1` and `_SC2` appear.

1 Input Columns

Column	Type	Meaning
<code>polity_name</code>	string	Name of the historical polity
<code>polity_region</code>	string	Geographic region
<code>leader_name</code>	string	Name of the leader
<code>leader_first_year</code>	int	First year of tenure (negative = BCE)
<code>leader_last_year</code>	int	Last year of tenure (negative = BCE)
<code>tenure</code>	int	Duration of tenure in years

2 Indicator Output Columns

Each indicator produces the following columns (replace {ind} with the indicator name, e.g. `sovereign_prediction`).

Column	Type	Meaning
<code>{ind}_prediction</code>	int	LLM prediction label
<code>{ind}_reasoning</code>	string	Step-by-step reasoning text
<code>{ind}_confidence</code>	int (1–100)	LLM self-reported confidence score
<code>{ind}_SC1</code>	int / null	Prediction from sample 1 (initial call)
<code>{ind}_SC2</code>	int / null	Prediction from sample 2
<code>{ind}_SC3</code>	int / null	Prediction from sample 3 (with <code>--n-samples 2</code>)
<code>{ind}_verified</code>	int / null	Majority-vote prediction across all samples; null if no consensus
<code>{ind}_agreement</code>	float (0–1)	Fraction of samples agreeing with the majority label
<code>{ind}_uncertainty</code>	string	none = all agree; low = partial; high = all differ

Indicator Label Definitions

Indicator	Values	Label Meanings
constitution	0, 1, 2	0 = no written law; 1 = code of law (governs subjects only); 2 = full constitution (written law + governance rules + limits on authority)
sovereign	0, 1	0 = colony, vassal, tributary, or protectorate; 1 = sovereign domestic governance
federalism	0, 1	0 = unitary state; 1 = sovereignty divided between central and local units (confederations, leagues, composite monarchies)
collegiality	0, 1	0 = single actor dominates decision-making; 1 = decisions genuinely shared by a constituted body
petition	0, 1	0 = petitioning is rare, ineffective, or unrecorded; 1 = petitions are a regular and institutionalized feature of political life
assembly	0, 1, 2, 3	0 = none; 1 = small advisory council (institutionalized); 2 = legislature with policymaking or selection role; 3 = popular assembly (most citizens or chosen by lot)
symbolism	0, 1, 2, 3	<i>Non-monotonic — higher code does not mean more power.</i> 0 = plain (ruler little distinguished from others); 1 = decorated (impressive but officeholder understood as mortal); 2 = deified (holder regarded as divine or quasi-divine); 3 = ceremonial (trappings so extensive they constrain power)
entry	0–10, 99	How the executive came to power: 0 = force (coup, rebellion, conquest); 1 = foreign power appointment; 2 = ruling party appointment (one-party system); 3 = royal council appointment; 4 = hereditary succession; 5 = military appointment; 6 = legislature appointment; 7 = head of state appointment; 8 = head of government appointment; 9 = direct popular election; 10 = other (e.g., clerical bodies such as College of Cardinals); 99 = unknown

Indicator	Values	Label Meanings
exit	0–14, 99	Circumstances of departure: 0 = voluntary retirement or abdication (not due to ill health); 1 = other regular exit (term limits, electoral defeat); 2 = regular transition to another office; 3 = died on campaign in civil war (disease or accident); 4 = died on campaign in foreign war (disease or accident); 5 = died of natural causes; 6 = retired due to ill health; 7 = suicide; 8 = deposed by domestic actors; 9 = assassinated or forced suicide; 10 = killed in battle in civil war; 11 = killed in battle in foreign war; 12 = irregular transition to another office; 13 = deposed by foreign state; 14 = still in office at end of observation period; 99 = unknown

3 Constitution-Specific Columns

Column	Type	Meaning
constitution_document_name	string	Semicolon-separated names of referenced constitutional documents, or "N/A"
constitution_year	string	Semicolon-separated years corresponding to those documents
constitution_document_types	string	Semicolon-separated type codes (1 or 2) matching the document names

4 Checks Sub-Indicator Columns

The `checks` indicator is split into nine binary sub-indicators coding whether that actor *actually* checks the executive (0 = does not check; 1 = does check). Each sub-indicator generates the same column set as any other indicator (`_prediction`, `_reasoning`, `_confidence`, `_SC1`, `_SC2`, `_SC3`, `_verified`, `_agreement`, `_uncertainty`).

Sub-indicator prefix	Actor
checks_local	Local or constituent territorial/kin-based units
checks_military	Armed forces or a warrior estate
checks_clergy	Religious authorities

Sub-indicator prefix	Actor
checks_aristocracy	Hereditary or titled upper stratum
checks_bourgeoisie	Urban commercial or capital-holding class
checks_bureaucracy	Professional administrative apparatus
checks_judiciary	Independent adjudicative bodies
checks_assembly	Deliberative representative or popular body
checks_council	Institutionalized elite advisory council

5 Elections Columns (Downstream)

Produced by `pipeline/post_processing.py`, run separately after the main pipeline. Rows where `assembly_prediction` $\neq 2$ receive a pass-through label of 0 with no API call.

Column	Type	Meaning
<code>elections_prediction</code>	int	0 = not elected (or pass-through when <code>assembly</code> $\neq 2$); 1 = elected, no organised factions; 2 = competitive elections (organised factions or parties)
<code>elections_reasoning</code>	string / null	Reasoning text; null for pass-through rows
<code>elections_confidence</code>	int / null	Confidence 1–100; null for pass-through rows
<code>elections_SC1</code>	int / null	Prediction from sample 1
<code>elections_SC2</code>	int / null	Prediction from sample 2
<code>elections_SC3</code>	int / null	Prediction from sample 3 (with <code>--n-samples 2</code>)
<code>elections_verified</code>	int / null	Majority-vote prediction; null if no consensus
<code>elections_agreement</code>	float (0–1)	Fraction of samples agreeing with the majority label
<code>elections_uncertainty</code>	string	none / low / high